

Mathematics II (part 1)

CONTINUOUS PROBABILITY DISTRIBUTION



Ryan Fraser Kirwan

CONTINUOUS PROBABILITY DISTRIBUTION

INTRODUCTION

Continuous probability distribution

Introduction

Discrete random variable:

- mean $\mu = \sum x_i P(X = x_i)$
- variance $\sigma^2 = \sum (x_i - \mu)^2 P(X = x_i)$
- standard deviation $\sigma = \sqrt{\sigma^2}$

Binomial random variable $X \sim B(n, p)$

- Probability distribution

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$$

- mean $\mu = np$
- variance $\sigma^2 = np(1-p)$

Key takeaways from previous lecture

What are we going to learn?

Continuous space

- Continuous random variables
- Understanding the concept

Density functions

- Recap with Histograms
- Discrete and Continuous

Different distributions

- Normal random variable $X \sim N(\mu, \sigma^2)$
- Z-tables

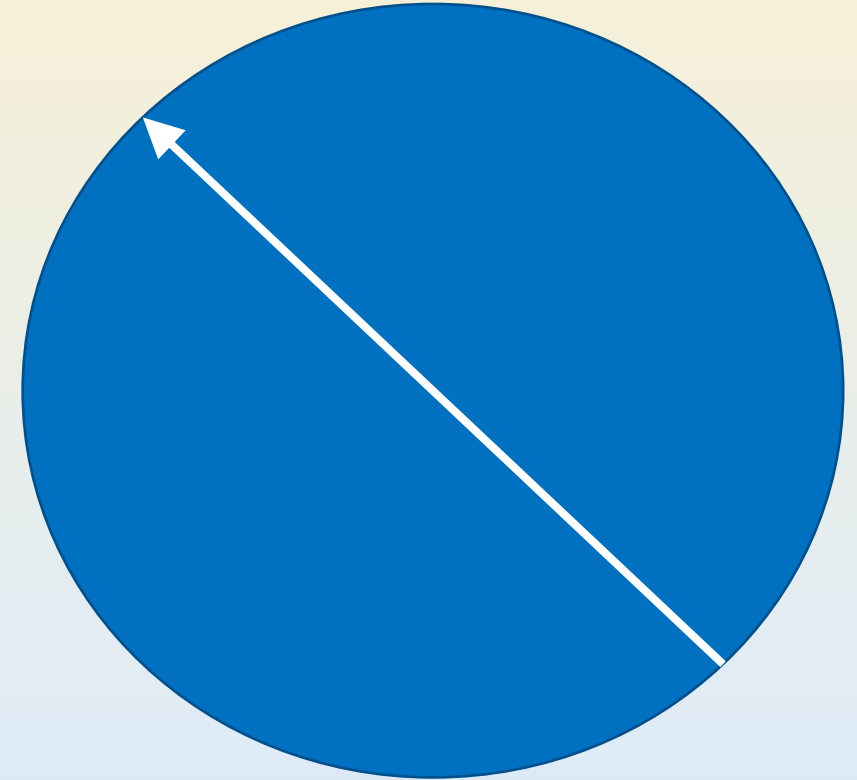
CONTINUOUS PROBABILITY DISTRIBUTION

CONTINUOUS SPACES

Continuous spaces

Spinning the arrow

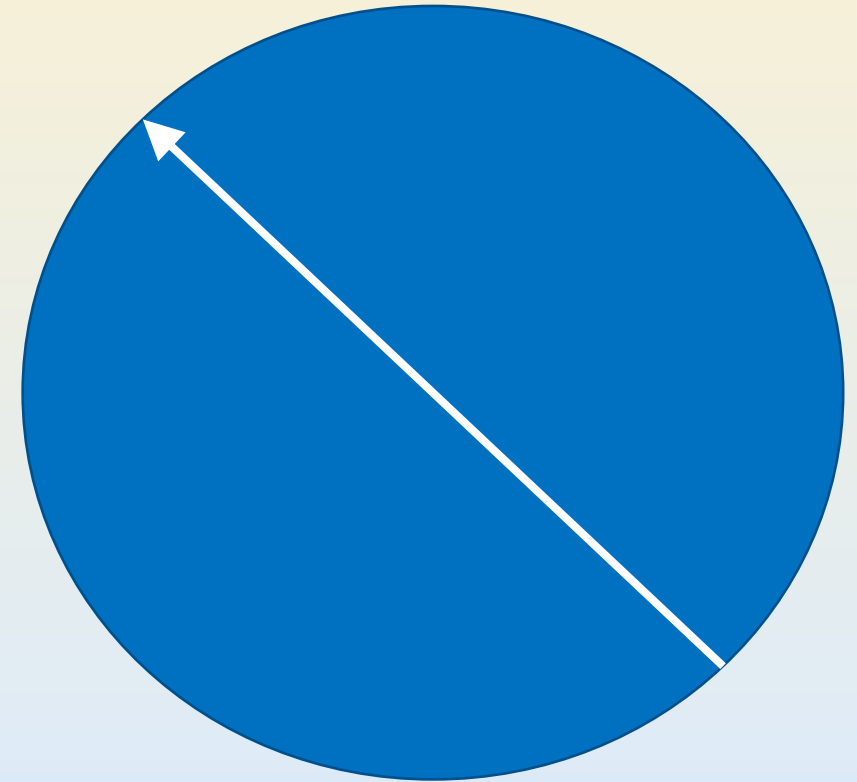
- We all know that the angles go from 0 to 360
- What is the probability that if I spin the arrow, it will take the **precise angle** of 179.9999998 degrees?
- **Precisely the angle** 179.9999998 Not 179.9999999 or 179.9999997?
(assume it is uniform, equally likely to be any angle)



Continuous spaces

Spinning the arrow

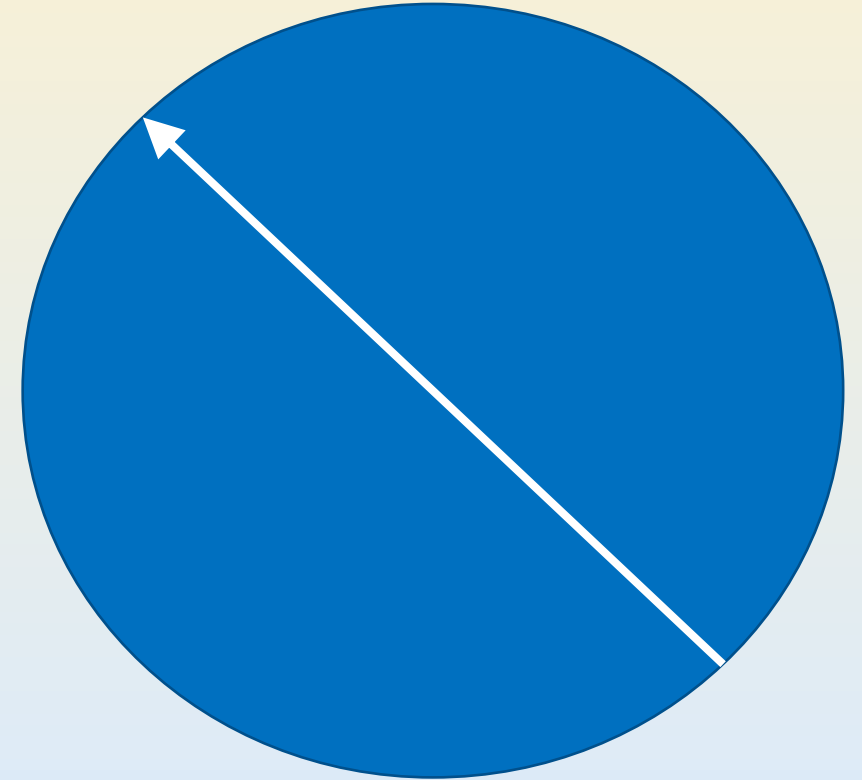
- What is the probability that if I spin the arrow, it will take the **precise angle** of 179.9999998 degrees?
- It **is very unlikely** to get a precise angle with the exact decimals.
- It **is actually Zero** to get any exact value in such continuous space.
- It sounds strange (**counter intuitive**), but it makes sense in continuous spaces



Continuous spaces

Spinning the arrow

- But does this mean, the arrow stops nowhere?
- Of course it stops...
- It could stop at 101.345354363234436431344
- It **is very unlikely** to get a precise angle with the exact decimals.

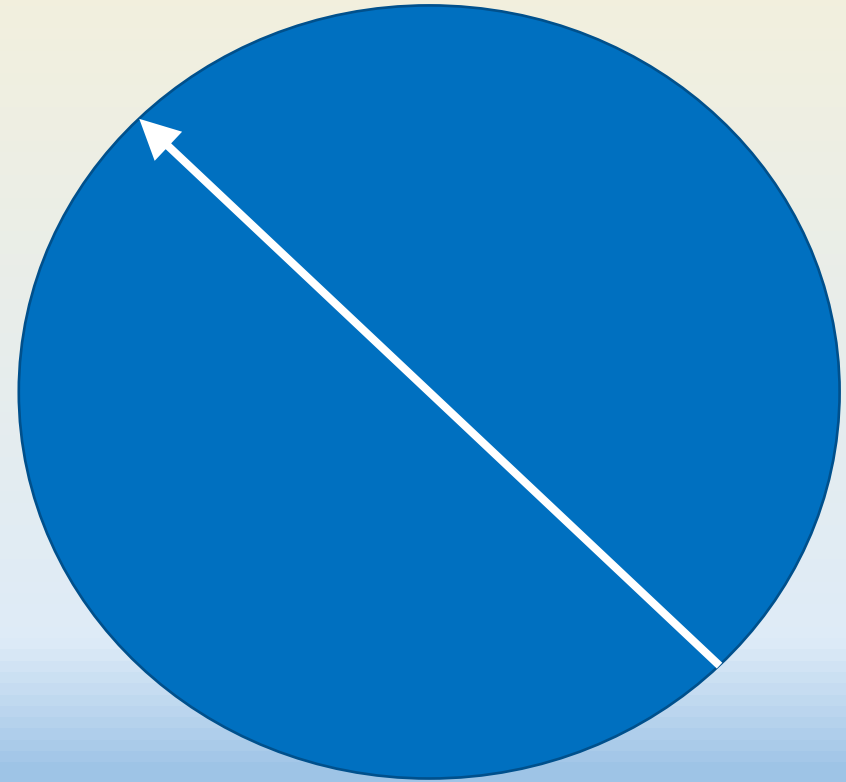


Continuous spaces

Spinning the arrow: Examples

Now, what is the probability that the arrow will stop between 0 and 180 degrees?

$$P(0 \leq x \leq 180) ?$$



Continuous spaces

Spinning the arrow: Examples

Now, what is the probability that the arrow will stop between 0 and 180 degrees?

$$P(0 \leq x \leq 180) ?$$

$$\frac{(180-0)}{360} = 1/2 = 0.5$$

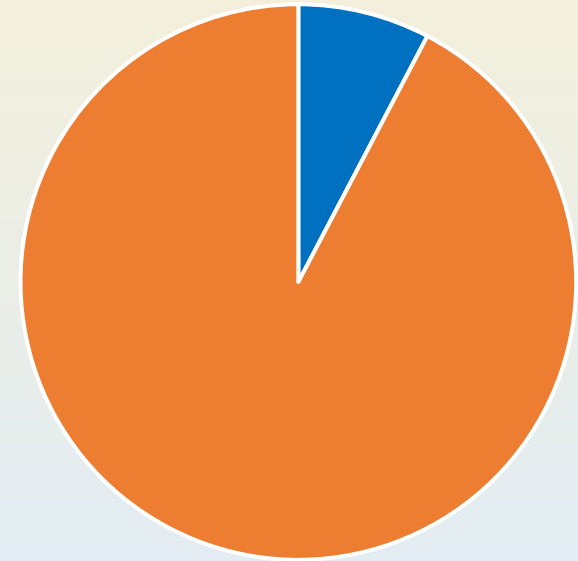


Continuous spaces

Spinning the arrow: Examples

Now, what is the probability that the arrow will stop between 260 and 290 degrees?

$$P(260 \leq x \leq 290)?$$



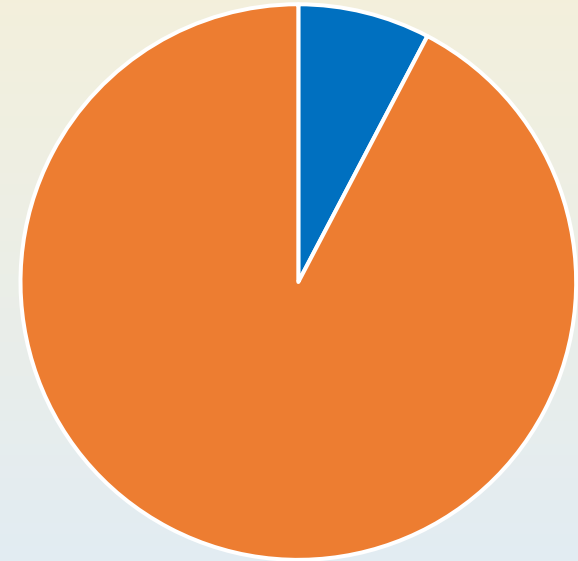
Continuous spaces

Spinning the arrow: Examples

Now, what is the probability that the arrow will stop between 260 and 290 degrees?

$$P(260 \leq x \leq 290)?$$

$$\frac{290 - 260}{360} = 1/12$$

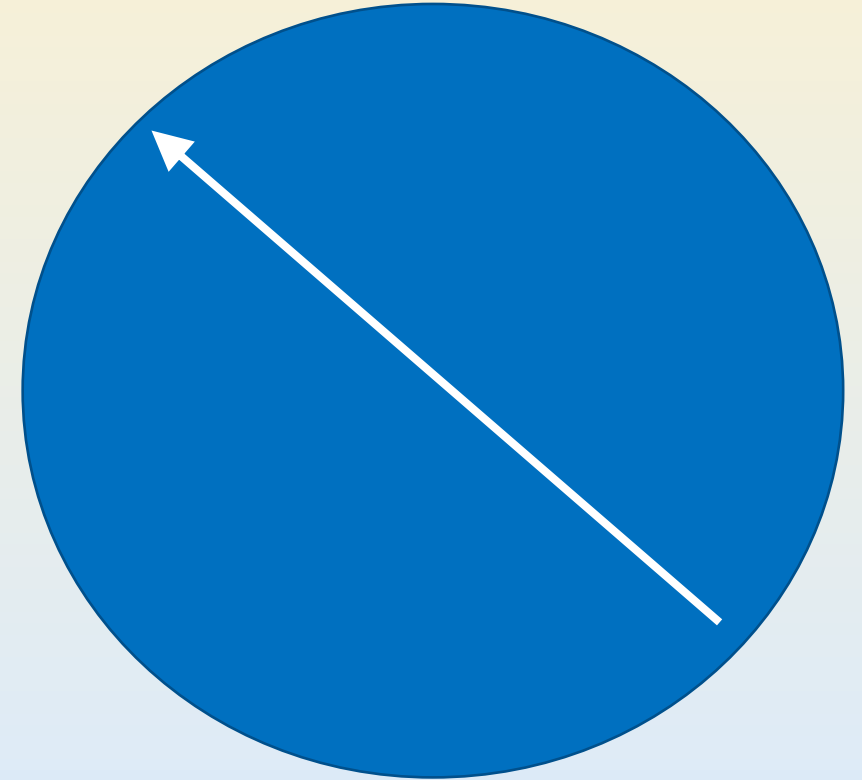


Continuous spaces

Spinning the arrow: Examples

Now, what is the probability that the arrow will stop between 179 and 180 degrees?

$$P(179 \leq x \leq 180)?$$



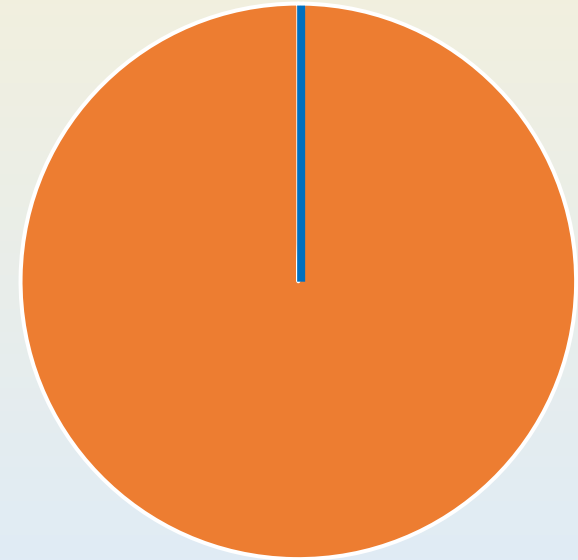
Continuous spaces

Spinning the arrow: Examples

Now, what is the probability that the arrow will stop between 179 and 180 degrees?

$$P(179 \leq x \leq 180)?$$

$$\frac{180-179}{360} = 1/360$$

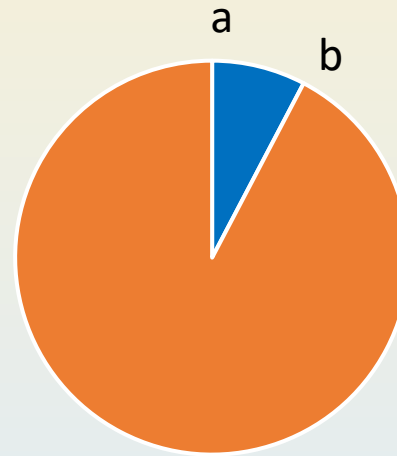


Continuous spaces

Spinning the arrow: Examples

For all the previous problems of intervals $[a,b]$ (when all possibilities are equally likely), it is solved by:

$$\frac{|b - a|}{360}$$

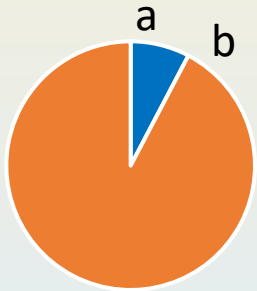


In other words, the probability is : $\frac{\text{size of requested interval}}{\text{size of the whole interval}}$

Continuous spaces

Spinning the arrow: Plotting Examples

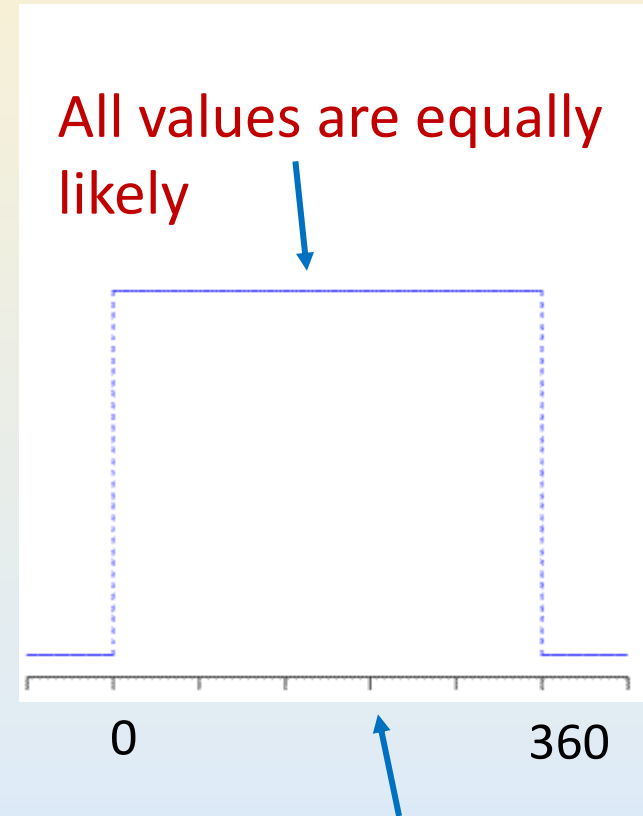
For all the previous problems of intervals $[a,b]$ (when all possibilities are equally likely), it is solved by:



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In other words, the probability is :

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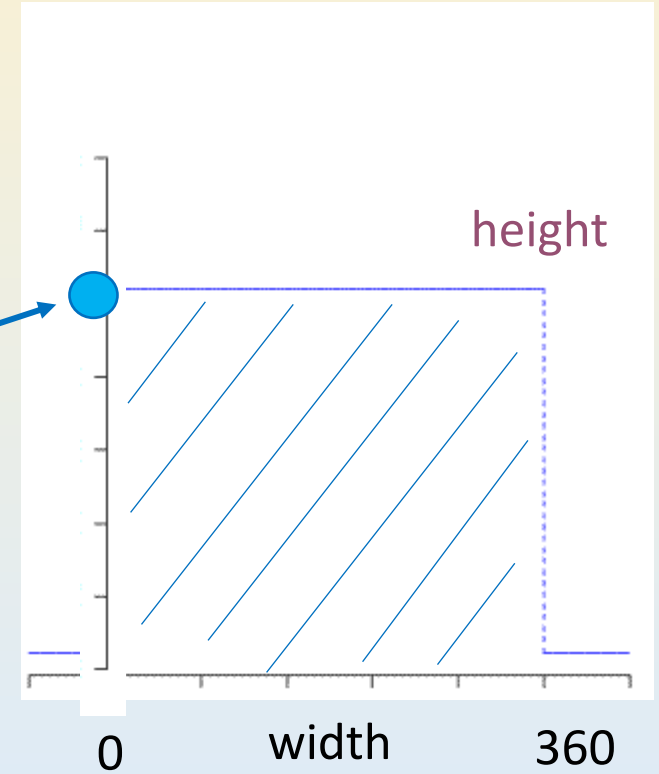


Angle can be anything between 0 and 360

Continuous spaces

Spinning the arrow: Plotting Examples

- Area under the curve is the **probability**, not **height**
- The area over all values = 1
 - $\text{area} = \text{height} * \text{width} = 1$
- What is the **height**? (value of the blue point)



Continuous spaces

Spinning the arrow: Plotting Examples

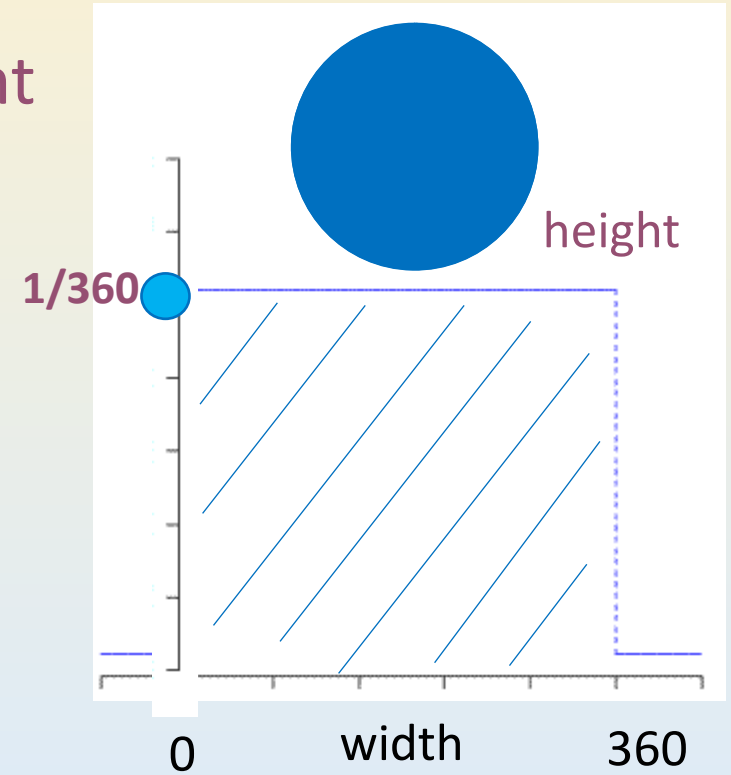
- Area under the curve is the **probability**, not **height**
- The area over all values = 1
 - $\text{area} = \text{height} * \text{width} = 1$
- What is the **height**? (value of the blue point)

Shaded area:

$$\text{height} = 1/360$$

$$\text{width} = 360$$

$$\text{area} = 1$$



CONTINUOUS PROBABILITY DISTRIBUTION

DENSITY FUNCTIONS

Density Functions

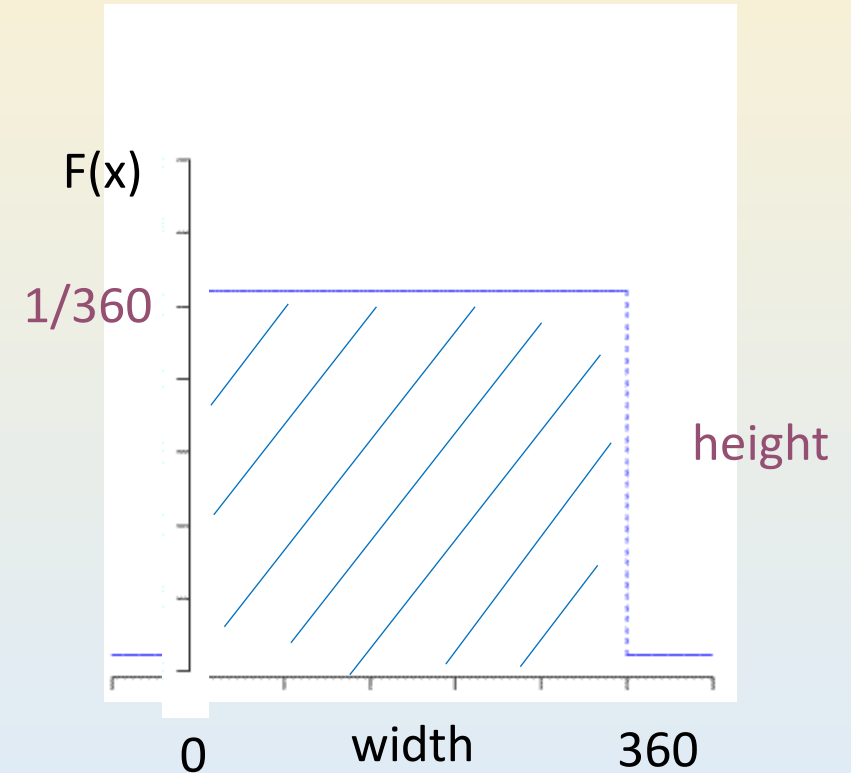
Understanding Probability Density Functions

This takes us to the concept of **Density**.

Density here is the probability for a continuous random value.

We represent the **height** by a density function $F(x)$

$F(x)$ in this example is $= \frac{1}{(360-0)} = \frac{1}{(360)}$



Density Functions

Understanding Probability Density Functions

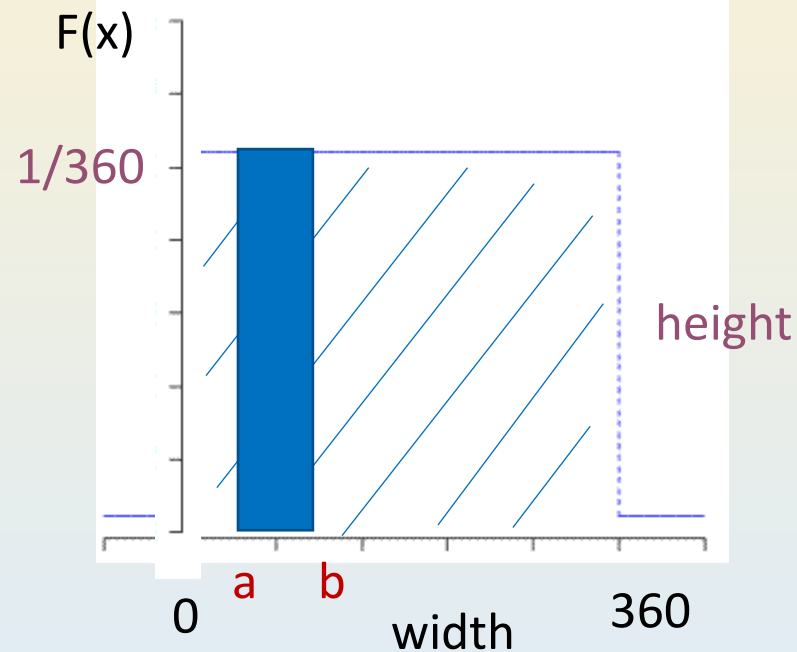
This takes us to the concept of **Density Function**.

$F(x)$ is density function in this example is $1/360$

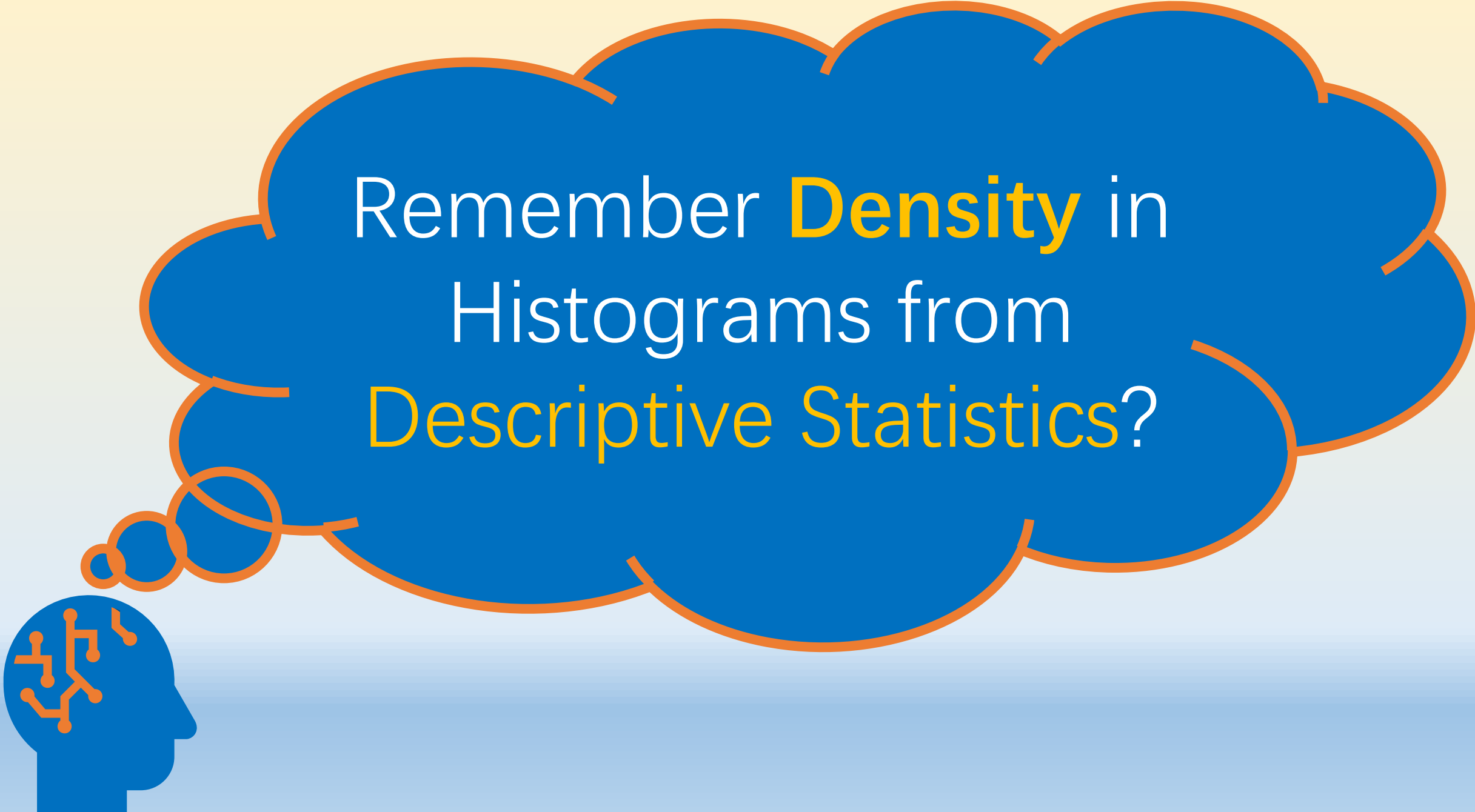
For **any interval** where $(a \leq x \leq b)$, the probability is the **area** under that **curve** (in this case, rectangle)

$$\text{Probability} = (b - a) * \text{height}$$

$$= (b - a) * \frac{1}{360} = \frac{b-a}{360}$$



Matches our
previous intuition of
size of interval over
whole interval

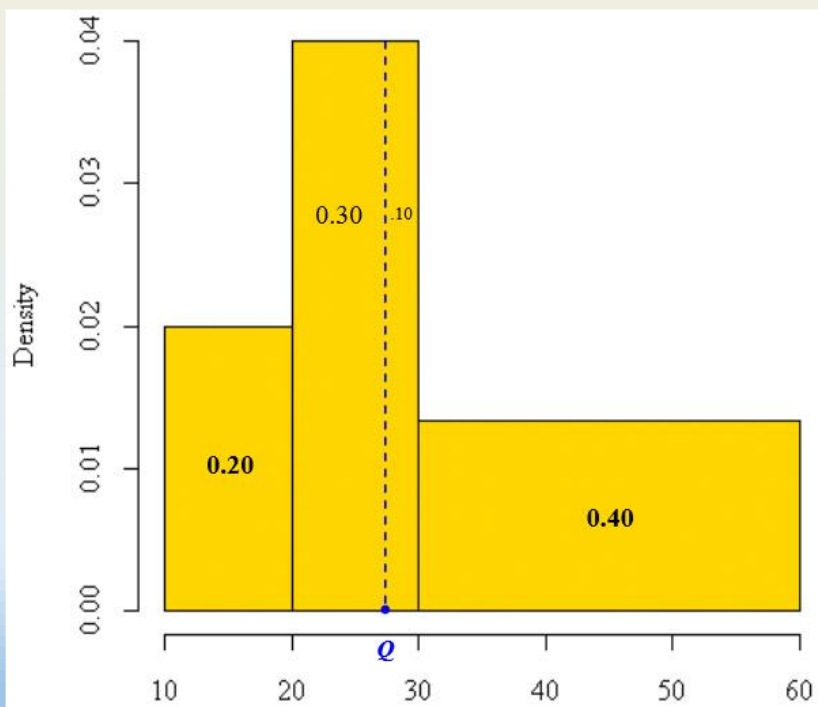


Remember **Density** in
Histograms from
Descriptive Statistics?

Density Functions

Histograms and Continuous variables

Recall in **Descriptive Statistics**, continuous data/variable are presented using frequency distribution or a **histogram**.



Class Interval	Frequency f_i	Relative Frequency $\frac{f_i}{n}$	Density (Rel Freq $\div$ Class Width)
[10, 20)	4	0.20	0.02
[20, 30)	8	0.40	0.04
[30, 60)	8	0.40	0.013

Density Functions

Density function: Examples

Let X be the **exact second** in which you were born—to infinite precision.

(assume it is uniform, equally likely to be any **second**)

What would be the $P(X)$?

Density Functions

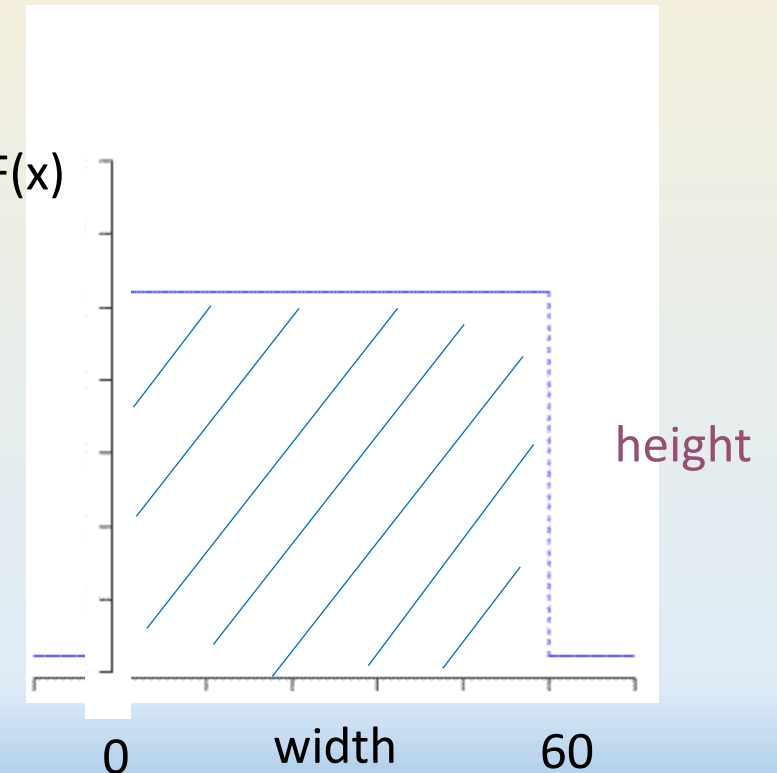
Density function: Examples

Let X be the **exact second** in which you were born—to infinite precision.

(assume it is uniform, equally likely to be any **second**) $F(x)$

What would be the $P(X)$?

What is the probability density function $F(X)$?



The possible value of seconds fall anywhere between 0 and 60

Density Functions

Density function: Examples

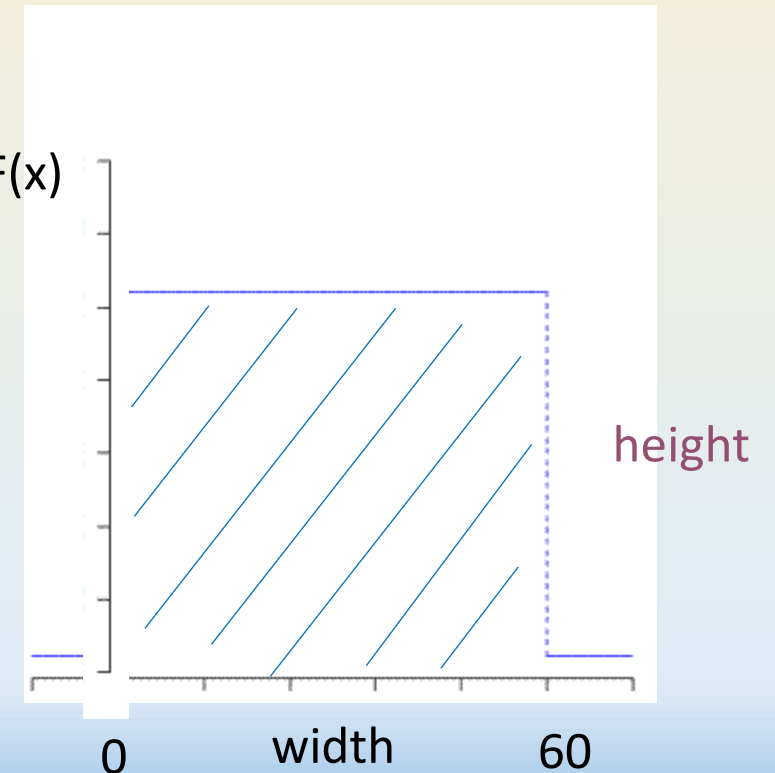
Let X be the **exact second** in which you were born—to infinite precision.

(assume it is uniform, equally likely to be any **second**) $F(x)$

What would be the $P(X)$? (**Zero for the exact value**)

What is the probability density function $F(X)$?

$1/60$

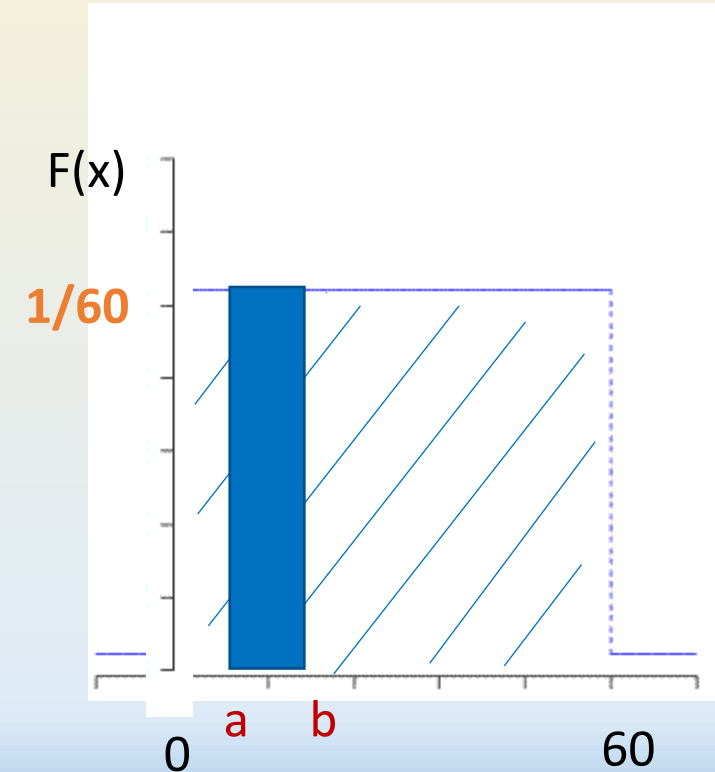


Density Functions

Density function: Examples

Let X be the exact **second** in which you were born
(assume it is uniform, equally likely to be any secs)

What is the probability that time is between the 10th second to 20th? $P(10 \leq x \leq 20)$?



Density Functions

Density function: Examples

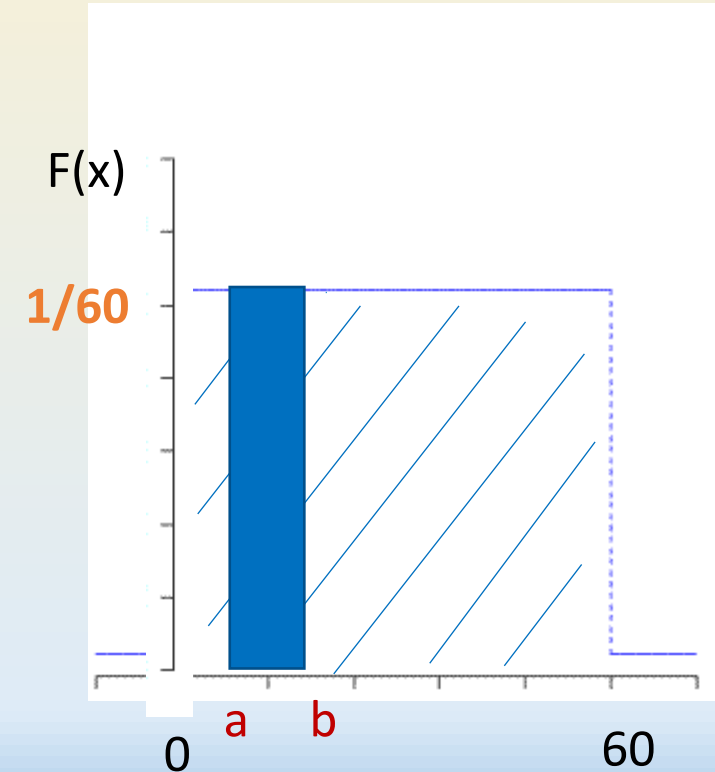
Let X be the exact **second** in which you were born
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What is the probability that time is between the
10th second to 20th? $P(10 \leq x \leq 20)$?

$$P(X) = (20 - 10)/60 = 10/60 = 1/6$$

i.e., Area of the **blue** rectangle

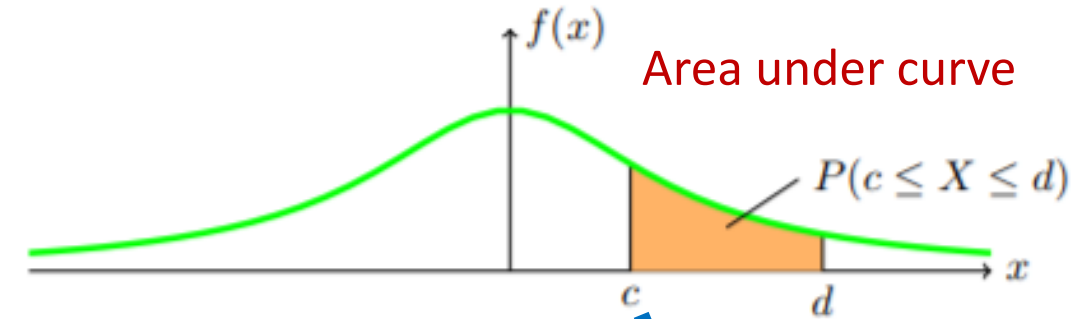
(**area under curve is the probability**)



Density Functions

Formal definition of Probability Density Functions

- **Continuous** random variables are described by probability distributions known as **probability density functions** (pdf).
- Definition: The **probability density function** of a **continuous** random variable X is a function $f(x)$ such that the area under the curve over an interval equals the probability of X in that interval.
- The **pdfs** we discussed so far (in previous slides) are for uniform distributions (where all outcomes are equally likely)



pdf and probability

Probability density function (pdf)

$$f(x) \geq 0; \quad P(c \leq x \leq d) = \int_c^d f(x) dx.$$

Density Functions

Formal definition of Probability Density Functions

Since there are **infinitely many values** (outcomes) for a continuous random variable, the probability for a **particular outcome** to occur is negligible; i.e.

$$P(X = x) = \int_x^x f(x) dx = 0$$

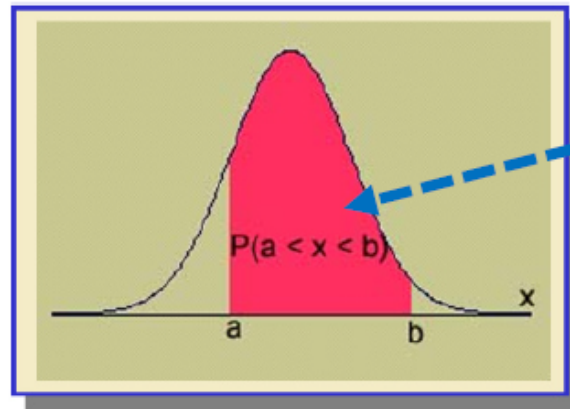
Density Functions

Formal definition of Probability Density Functions

Since there are **infinitely many values** (outcomes) for a continuous random variable, the **probability for a particular outcome to occur is negligible; i.e.**

$$P(X = x) = \int_x^x f(x)dx = 0$$

Therefore, for continuous random variable, we find probability for an interval of values (outcomes) to occur:



$$\begin{aligned} P(a < X < b) \\ &= P(a \leq X < b) \\ &= P(a < X \leq b) \\ &= P(a \leq X \leq b) \\ &= \int_a^b f(x)dx \end{aligned}$$

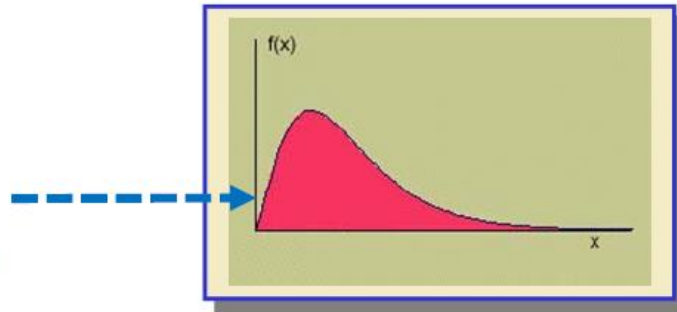
Density Functions

Compare terminologies and concepts with discrete distribution definition

Continuous random variables

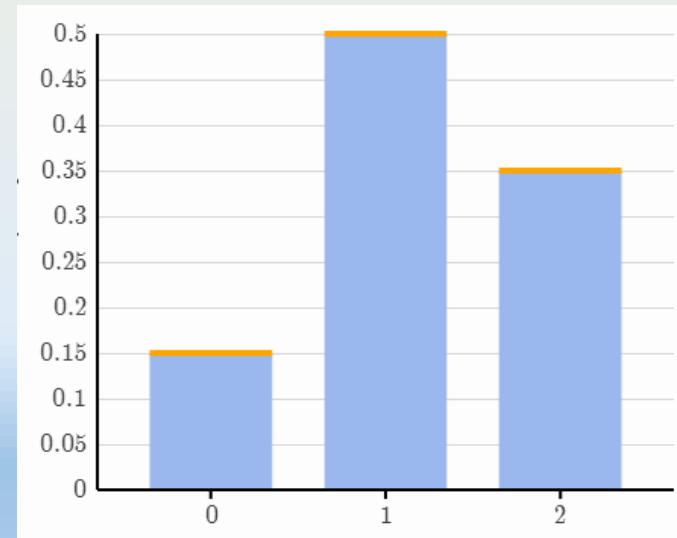
(i) $f(x) \geq 0$

(ii) $\int_{-\infty}^{\infty} f(x) = 1$
area under curve



Discrete random variable

$$P(X = x) \geq 0 \quad \text{and} \quad \sum P(X = x) = 1$$



Density Functions

Compare terminologies and concepts with discrete distribution definition

Discrete and Continuous probability distribution

Discrete random variable	Continuous random variable
(a) Expectation or mean $E(X) = \sum xP(X = x)$	(a) Expectation or mean $E(X) = \mu = \int_{-\infty}^{\infty} x f(x) dx$
(b) Variance $Var(X) = \sum (x_i - \mu)^2 P(X = x_i)$ or $Var(X) = E(X^2) - \mu^2$ where $E(X^2) = \sum x^2 P(X = x)$	(b) Variance $Var(X) = \sigma^2 = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx$ or $Var(X) = \sigma^2 = E(X^2) - \mu^2$ where $E(X^2) = \int_{-\infty}^{\infty} x^2 f(x) dx$

Similar to discrete random variables, continuous random variables also have mean and variance.

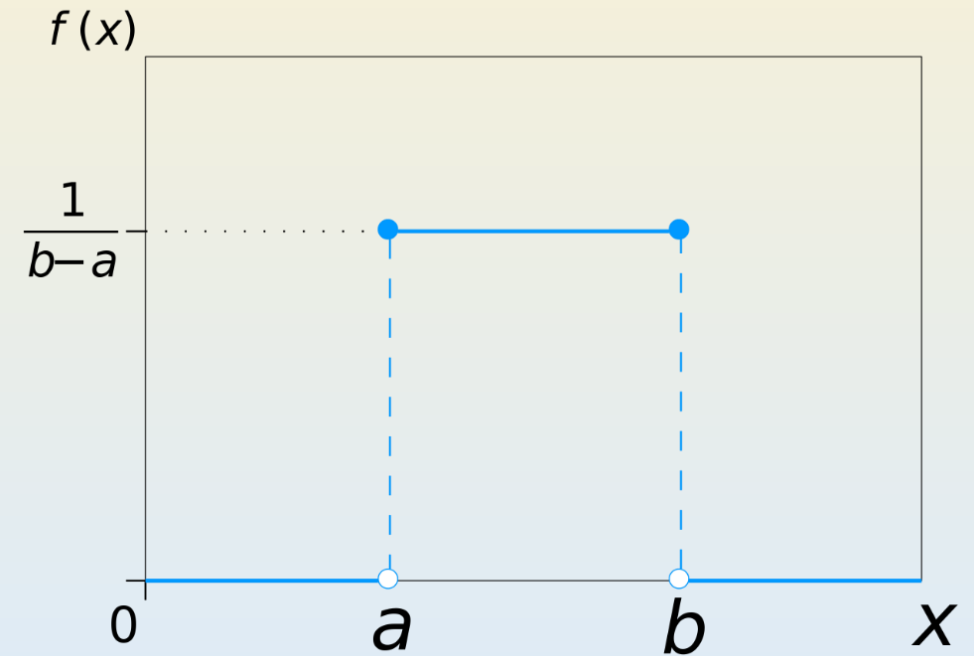
CONTINUOUS PROBABILITY DISTRIBUTION

DIFFERENT DISTRIBUTIONS

Different distributions

Uniform distribution

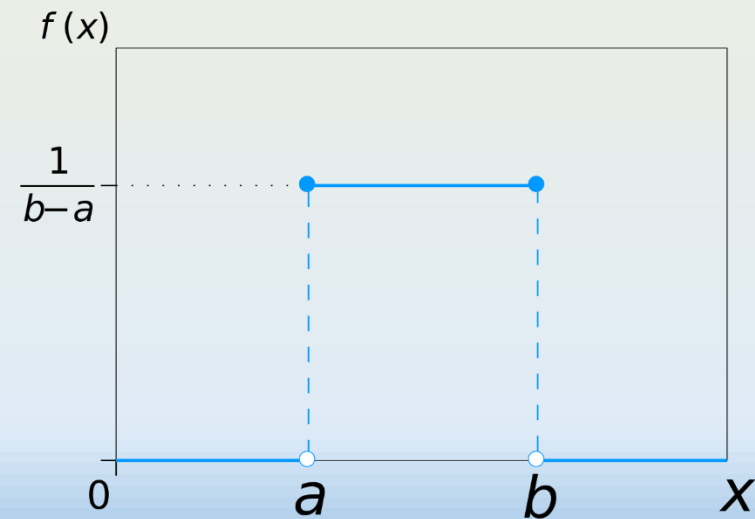
- The **continuous uniform distribution** or **rectangular distribution** is a family of **symmetric** probability distributions such that for each member of the family, all intervals of the same length on the distribution's support are **equally probable**. ([Wikipedia](#))
- The function is defined by the two parameters, a and b , which are its **minimum** and **maximum** values of the required interval.



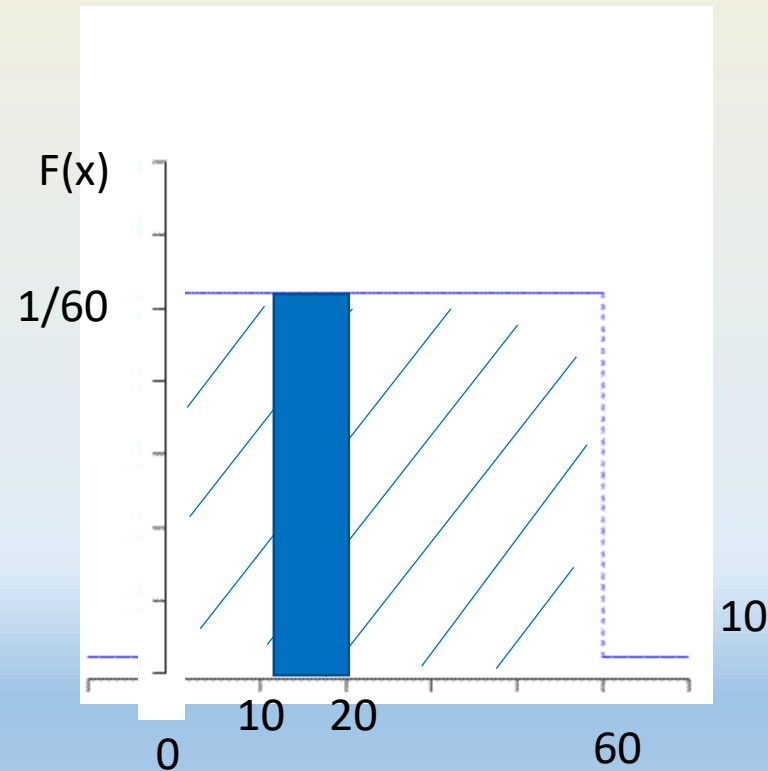
Different distributions

Uniform distribution Examples

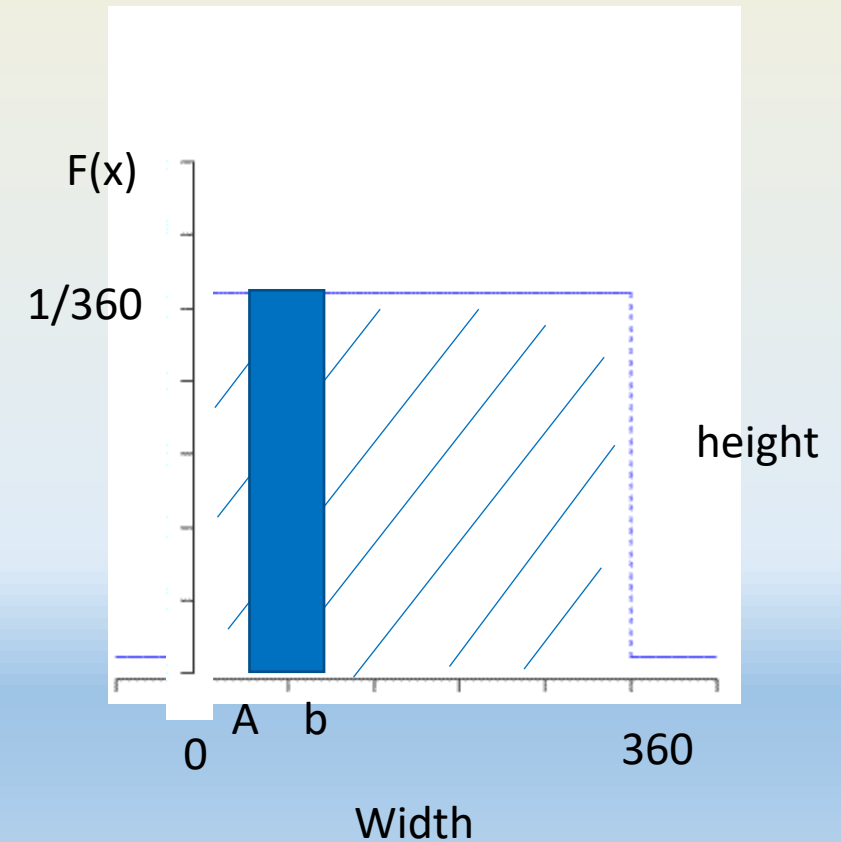
Uniform Distributions



Time of birth Problem



Spinning arrow problem

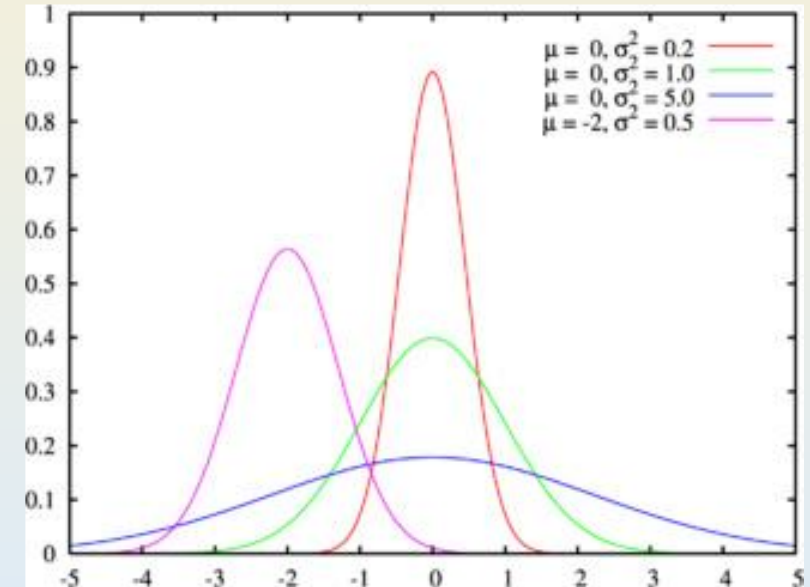


Different distributions

Normal distribution

Another **continuous random** variable is the **normal random variable** or **normal distribution** which is very useful to model many random phenomena in the real world.

Definition: A normal random variable or normal distribution X , written as $X \sim N(\mu, \sigma^2)$, is a **continuous random** variable having **the shown** probability density function.



Different distributions

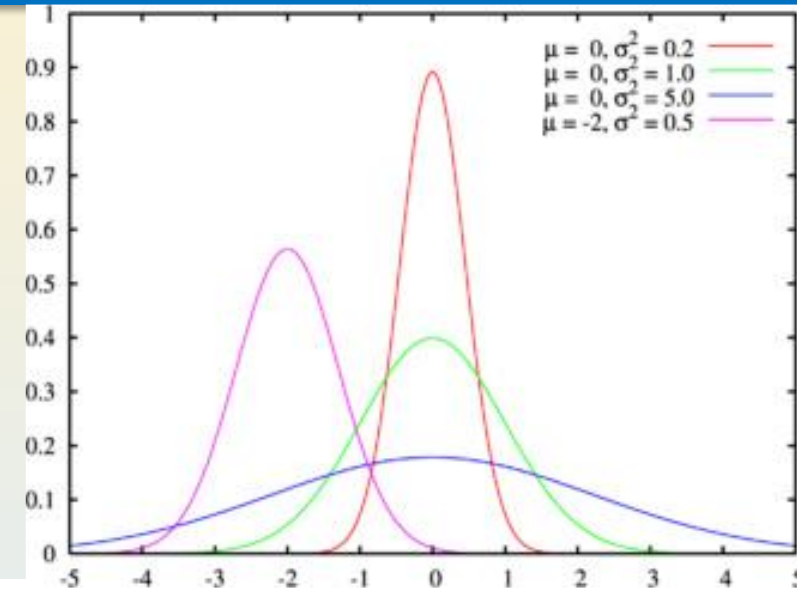
Normal distribution

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Definition: A normal random variable or normal distribution X , written as $X \sim N(\mu, \sigma^2)$, is a **continuous random** variable having **the shown** probability density function.

The parameters μ and σ^2 are the **mean** and **variance** of the normal random variable.

Note that **different** values of **mean** and **variance** represent **different normal distributions**.



Probability density function (pdf)

$$f(x) \geq 0; \quad P(c \leq x \leq d) = \int_c^d f(x) dx.$$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$$-\infty < x < \infty$$



DO WE **NEED** TO USE
THESE EQUATIONS
TO GET THE
PROBABILITY?

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$$-\infty < x < \infty$$

Probability density function (pdf)

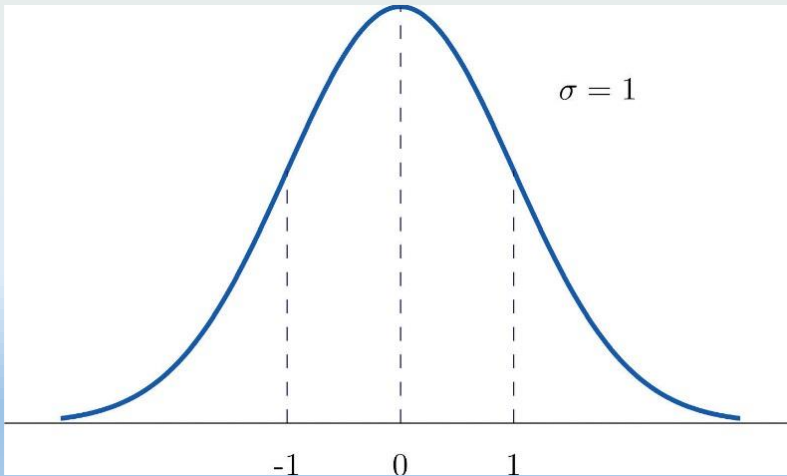
$$f(x) \geq 0; \quad P(c \leq x \leq d) = \int_c^d f(x) dx.$$

Different distributions

Standard normal probability distribution

It is **DIFFICULT** to calculate probabilities (area under curve) of a normal distribution directly. To simplify, all normal distributions can be transformed into a standard normal.

Definition: A standard normal written as $Z \sim N(0, 1)$, has a mean of 0 and variance of 1, and has the following pdf:



$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} \quad -\infty < x < \infty$$

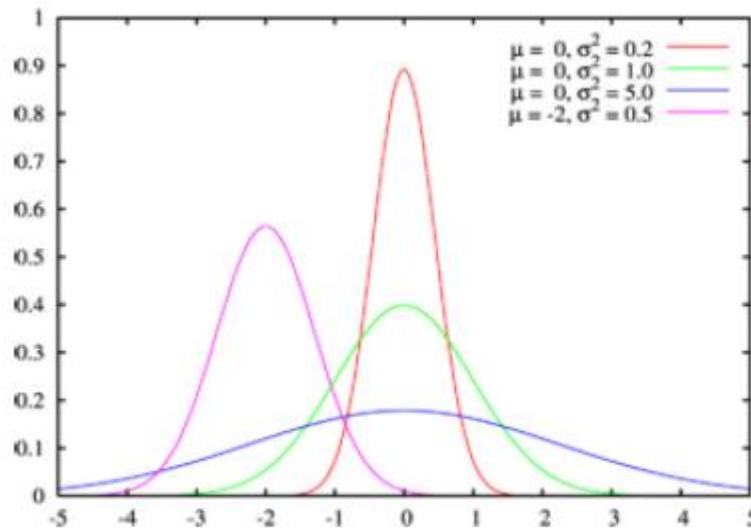
Different distributions

From Normal to Standard normal probability distribution

General normal distributions X transforming to standard normal Z , how?

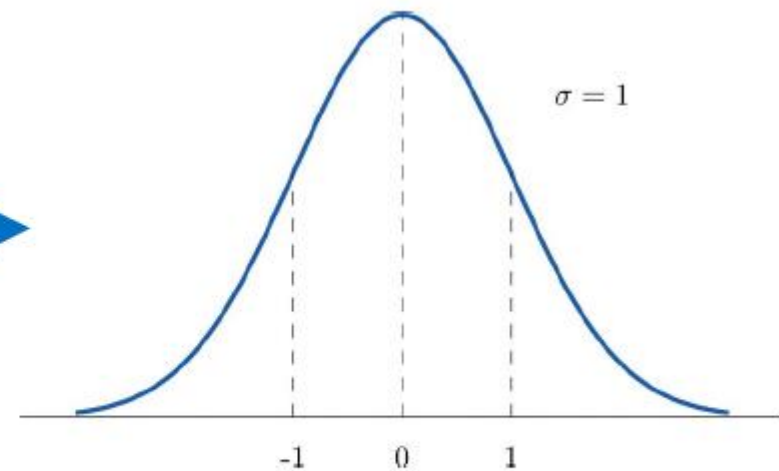
$$X \sim N(\mu, \sigma^2)$$

$$Z \sim N(0, 1)$$



$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$$-\infty < x < \infty$$



$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} \quad -\infty < x < \infty$$

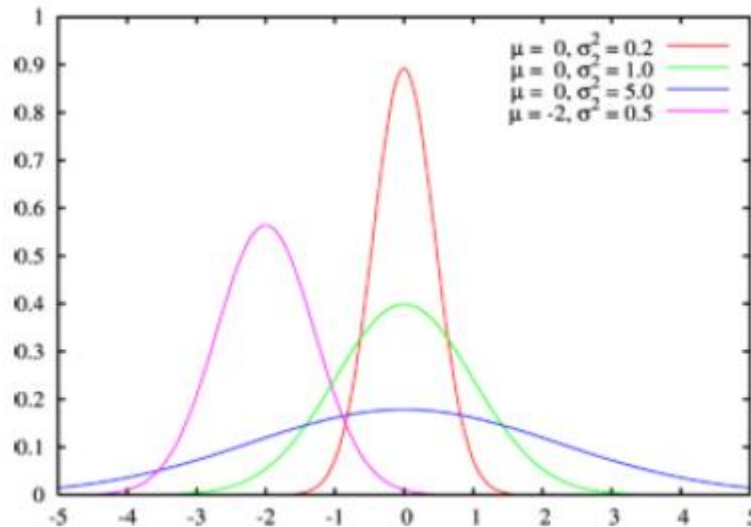
Different distributions

From Normal to Standard normal probability distribution

General normal distributions X transforming to standard normal Z , how?

$$X \sim N(\mu, \sigma^2)$$

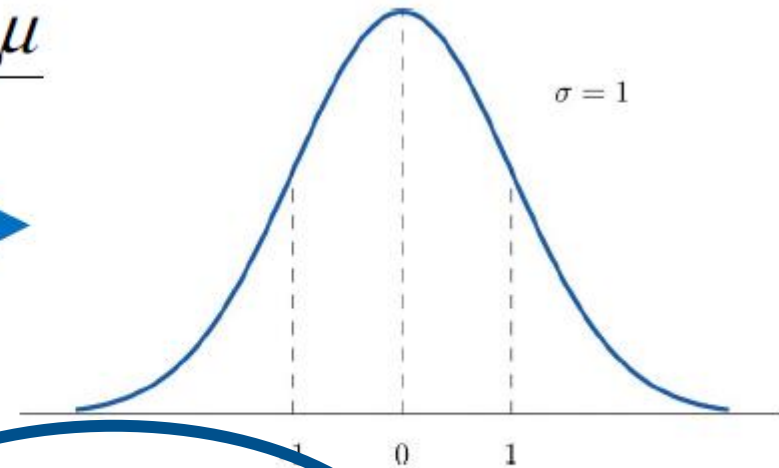
$$Z \sim N(0, 1)$$



$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$$-\infty < x < \infty$$

$$Z = \frac{X - \mu}{\sigma}$$



$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} \quad -\infty < x < \infty$$

STANDARD NORMAL DISTRIBUTION

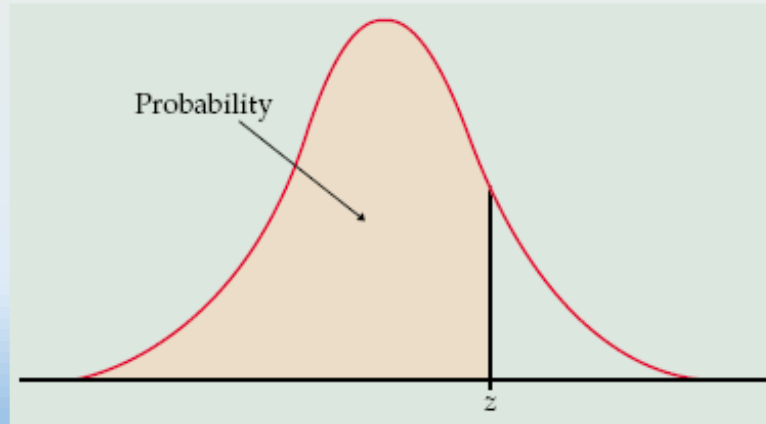
Z-TABLES

Z-tables

Introduction

Unfortunately, it is still difficult to calculate probabilities of a standard normal. So, we need to rely **on a table** of precomputed probabilities called **Z-table**.

$$P(Z < z)$$



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986

Z-tables

Example 1

An example on the use of Z-table to find probability.

Example: Find $P(Z \leq 1.36) = ?$

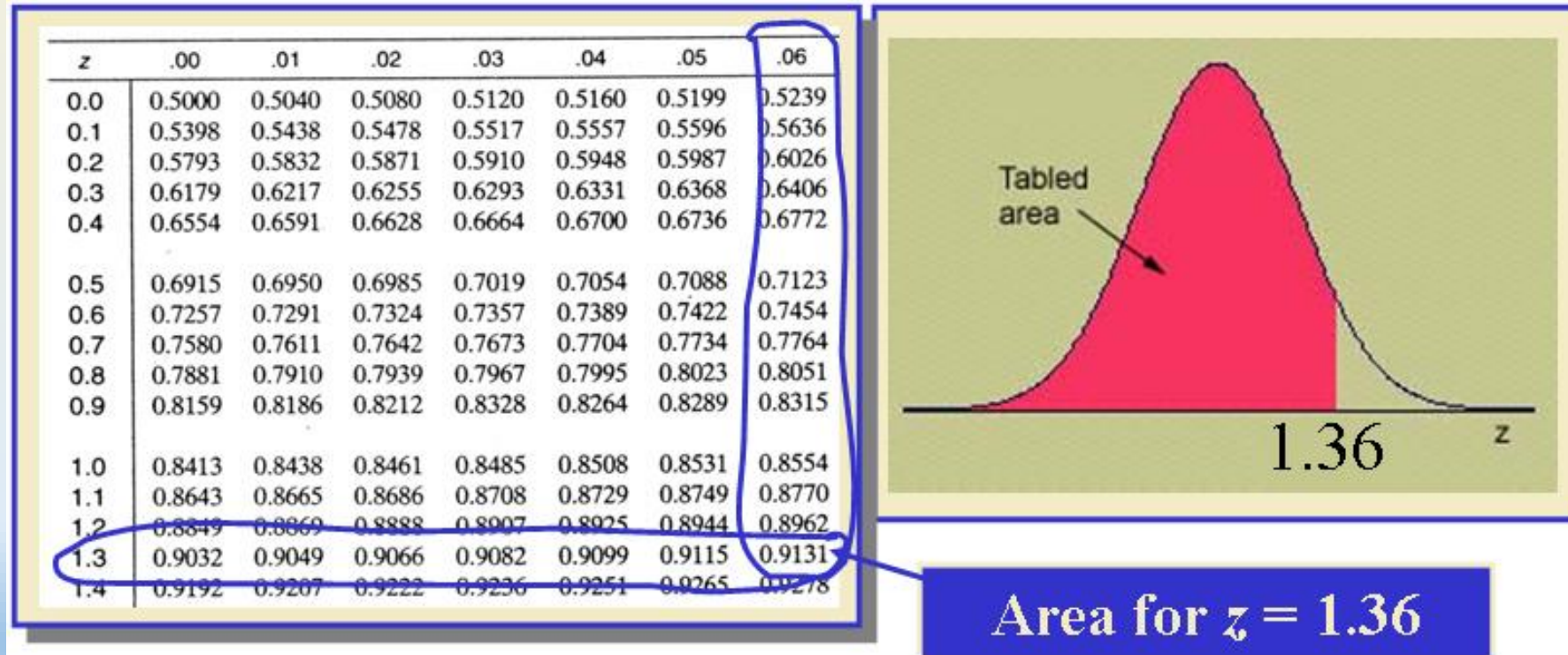
z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
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2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986

Z-tables

Example 1

An example on the use of Z-table to find probability.

Example: Find $P(Z \leq 1.36) = ?$



From Z-table above, $P(Z \leq 1.36) = 0.9131$

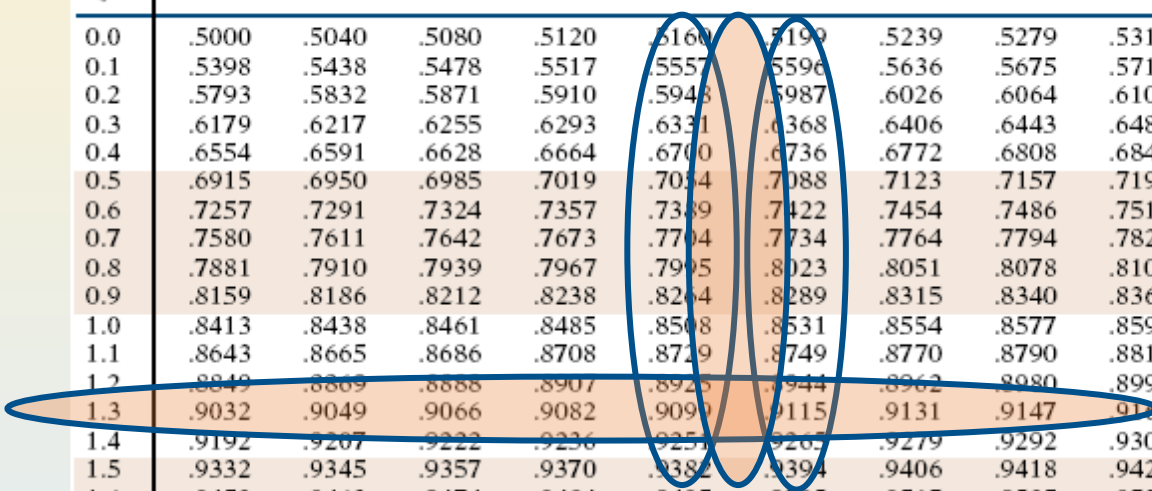
Z-tables

Example 2

An example on the use of Z-table to find probability.

Example: Find $P(Z \leq 1.345) = ?$

0.9107



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986

Z-tables

Example 3

An example on the use of Z-table to find probability.

Example: Find $P(Z > 1.11) = ?$

First find $P(Z \leq 1.11) = ?$

Z-tables

Example 3

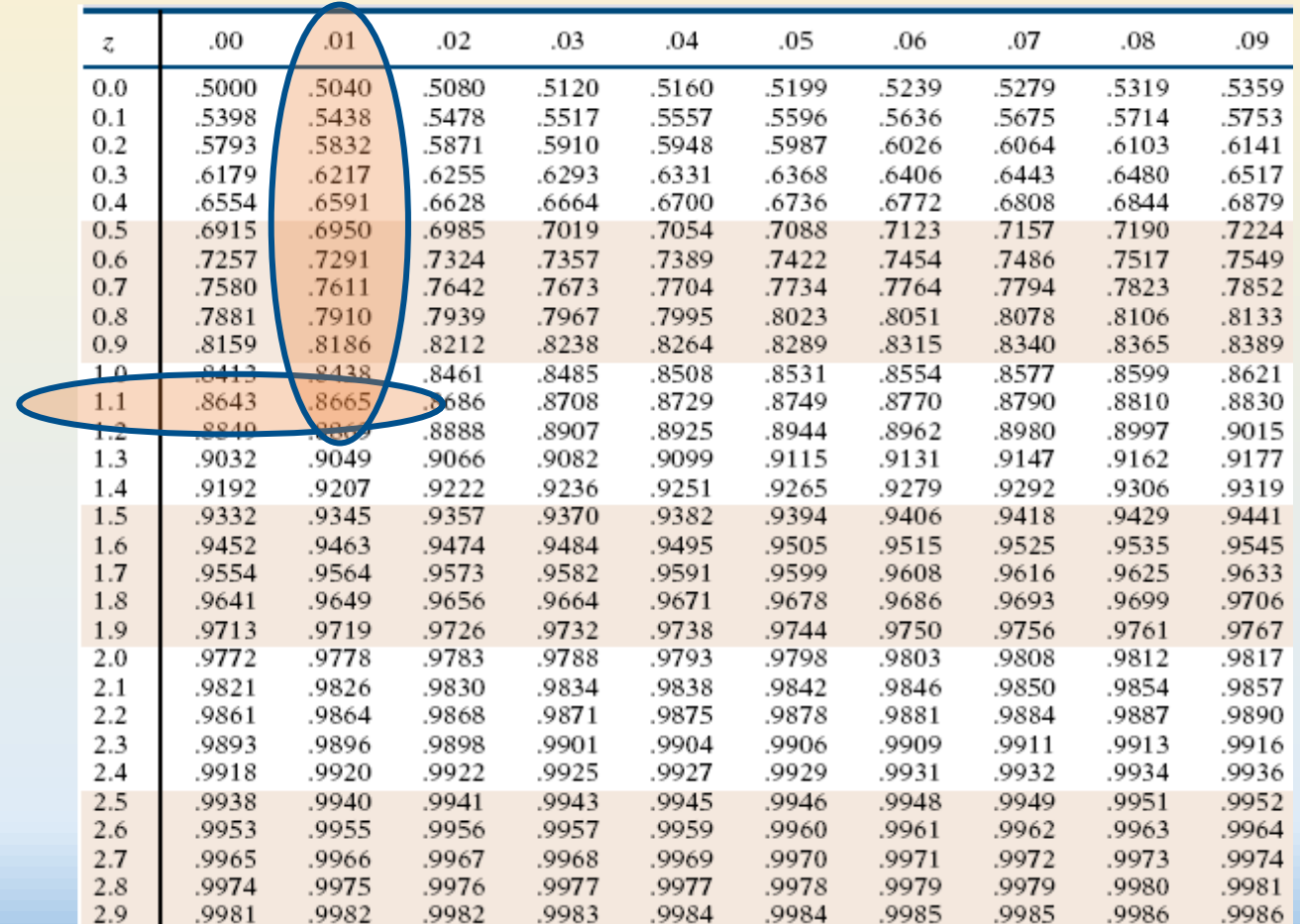
An example on the use of Z-table to find probability.

Example: Find $P(Z > 1.11) = ?$

(first find)

$$P(Z \leq 1.11) = 0.8665$$

$$P(Z > 1.11) = 1 - 0.8665 = 0.1335$$

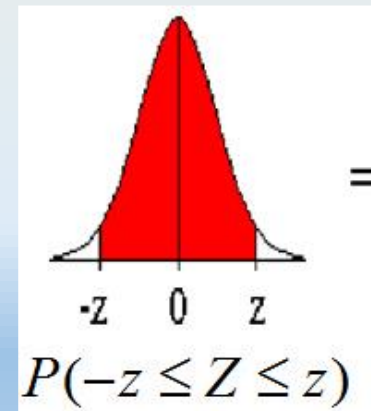
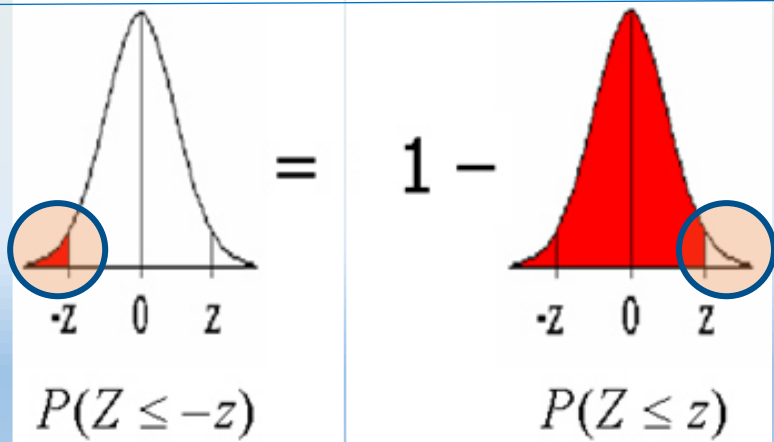
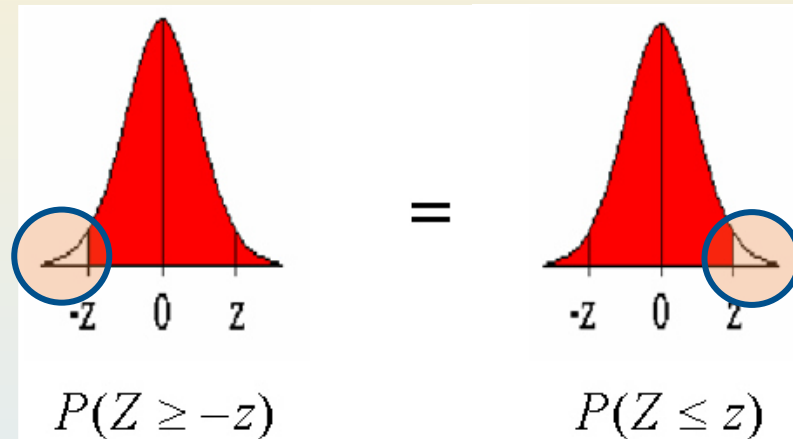
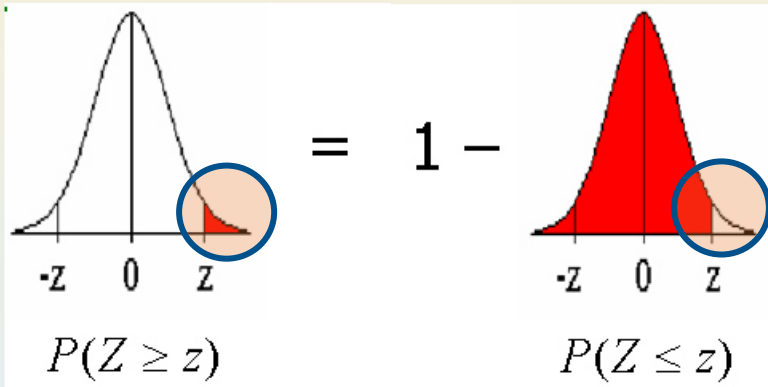


z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
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2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986

Z-tables

Symmetry Example

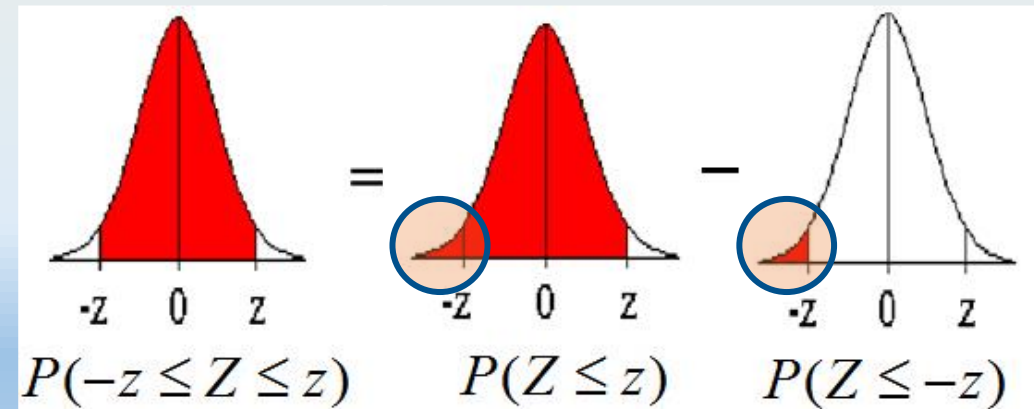
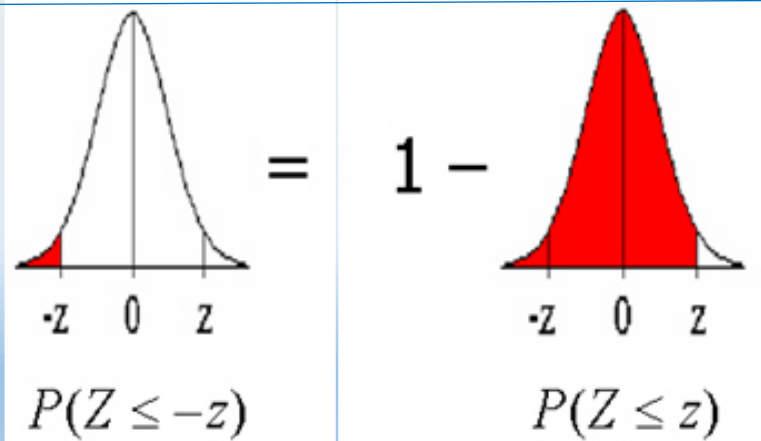
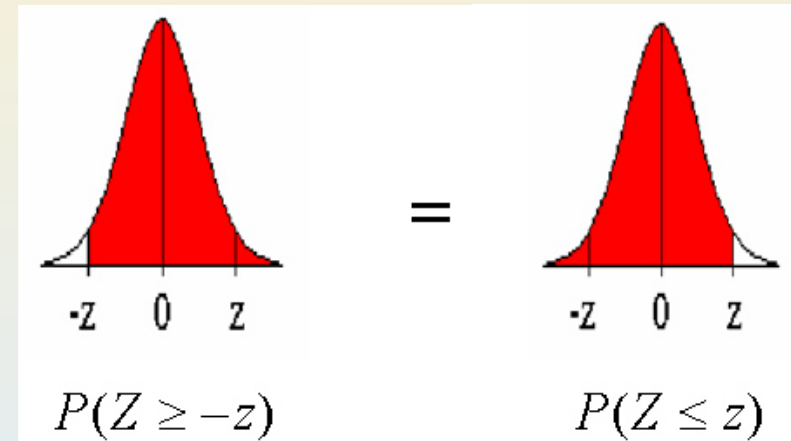
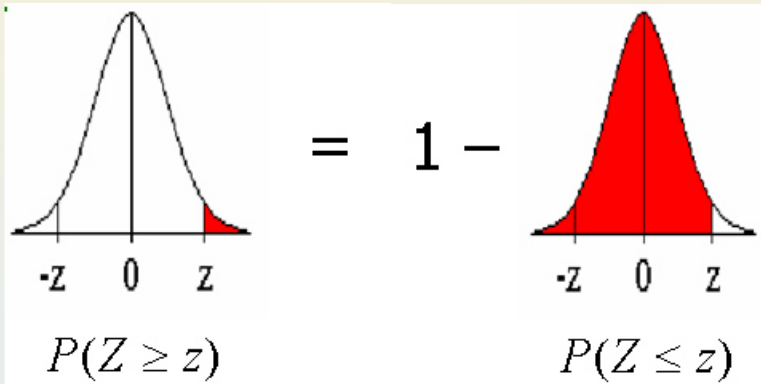
Although most Z-tables only tabulate $P(Z < z)$, it is not too difficult to determine other probabilities based on the fact that standard normal is **symmetrical about the mean**.



Z-tables

Symmetry Example

Although most Z-tables only tabulate $P(Z < z)$, it is not too difficult to determine other probabilities based on the fact that standard normal is **symmetrical about the mean**.



Reverse the question, you have the probability, what is the Z?

Example: For what value of z is $P(Z < z) = 0.95$?

Z-tables

Example 4

Reverse the question, you have the probability, what is the Z?

Example: For what value of z is $P(Z < z) = 0.95$?

From Z-table above, $z = 1.645$

Note: Since 0.95 is in between 0.9495 and 0.9505, we may choose the mid-point.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767

CONTINUOUS PROBABILITY DISTRIBUTION

SUMMARY

- ✓ Probability of any exact value in the continuous space is zero
- ✓ We use density curves to represent continuous probability distributions
- ✓ Uniform Continuous Probability distribution
- ✓ Normal Probability distribution
- ✓ Converting to Standard Normal Distribution
- ✓ Using the Z-Table